MATHEMATICS II FOR DATA SCIENCE
GA-5 SOLUTIONS MAY 2024

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1. Question

The marks obtained by Karthika, Romy and Farzana in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table $M2W5G1$.

	Quiz 1	Quiz 2	End sem
Karthika	40	50	60
Romy	20	50	50
Farzana	30	40	70

Table: M2W5G1

If $x_1\%$ of Quiz 1, $x_2\%$ of Quiz 2, and $x_3\%$ of End sem is taken to calculate the total marks $T_1(x_1, x_2, x_3)$, $T_2(x_1, x_2, x_3)$, $T_3(x_1, x_2, x_3)$ of Karthika, Romy, and Farzana, respectively, then choose the correct set of options.

- ☐ $T_2(x_1, x_2, x_3) = \frac{1}{100}(20x_1 + 50x_2 + 50x_3)$.
- ☐ $T_2(x_1, x_2, x_3) = \frac{1}{100}(40x_1 + 20x_2 + 30x_3)$.
- ☐ If $x_1 = 20\%$, $x_2 = 20\%$, and $x_3 = 60\%$, then $T_1(x_1, x_2, x_3) = 44$.
- ☐ If $x_1 = 20\%$, $x_2 = 20\%$, and $x_3 = 60\%$, then $T_1(x_1, x_2, x_3) = 54$.
- ☐ If $x_1 = 20\%$, $x_2 = 20\%$, and $x_3 = 60\%$, then Farzana obtained the highest total marks.
- ☐ $T_i(x_1, x_2, x_3)$ is a linear mapping from $\mathbb{R}^3$ to $\mathbb{R}$, for each $i = 1, 2, 3$.

1.1. Answers:

- $T_2(x_1, x_2, x_3) = \frac{1}{100}(20x_1 + 50x_2 + 50x_3)$.
- If $x_1 = 20\%$, $x_2 = 20\%$, and $x_3 = 60\%$, then $T_1(x_1, x_2, x_3) = 54$.
- If $x_1 = 20\%$, $x_2 = 20\%$, and $x_3 = 60\%$, then Farzana obtained the highest total marks.
- $T_i(x_1, x_2, x_3)$ is a linear mapping from $\mathbb{R}^3$ to $\mathbb{R}$, for each $i = 1, 2, 3$.

1.2. Solution

We can see that $T_2(x_1, x_2, x_3)$ refers to Romy's total marks, which is there in the second row of the given table. Romy scored 20 marks in Quiz 1, 50 marks in Quiz 2 and 50 marks in End sem. Hence, **option 1** is correct and **option 2** is incorrect.

$T_1(x_1, x_2, x_3)$ refers to Karthika's total marks, which is there in the first row of the given table, If $x_1 = 20\%,\ x_2 = 20\%\ and\ x_3 = 60\%$, then,

$$\begin{aligned} T_1(x_1, x_2, x_3) &= \frac{1}{100}(20 \times 40 + 20 \times 50 + 60 \times 60) \\ &= \frac{1}{100}(800 + 1000 + 3600) \\ &= \frac{1}{100}(5400) \\ &= 54 \end{aligned}$$

Hence, **option 4** is correct and **option 3** is wrong.

$T_3(x_1, x_2, x_3)$ refers to Farzana's total marks, which is there in the third row of the given table, If $x_1 = 20\%,\ x_2 = 20\%\ and\ x_3 = 60\%$, then,

$$\begin{aligned} T_3(x_1, x_2, x_3) &= \frac{1}{100}(20 \times 20 + 20 \times 50 + 60 \times 50) \\ &= \frac{1}{100}(600 + 800 + 4200) \\ &= \frac{1}{100}(5600) \\ &= 56 \end{aligned}$$

Notice that Farzana's marks in Quiz 1, Quiz 2 and End sem are greater than Romy's marks in Quiz 1, Quiz 2 and End sem respectively. Therefore, Farzana's total marks will be more than Romy's total marks. From the above calculations, we can see that Farzana's total marks is more than Karthik's total marks. Hence, **option 5** is correct.

Option 6 is correct. We send a vector with Quiz 1, Quiz 2 and End sem marks as input and we get the total marks as output. Therefore it is a linear mapping from $\mathbb{R}^3$ to $\mathbb{R}$.

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2. Question

A molecule of carbon dioxide (CO_2) consists of one carbon atom and two oxygen atoms. A molecule of water consists of two hydrogen atoms and one oxygen atom, whereas a molecule of oxygen consists of two oxygen atoms. One molecule of glucose consists of six atoms of carbon, twelve atoms of hydrogen, and six atoms of oxygen.

In the presence of sunlight, the leaves of trees take carbon dioxide and water to produce glucose and oxygen. The chemical equation corresponding to this chemical reaction (R) is given below.

$$x_1CO_2 \ + \ x_2H_2O \longrightarrow x_3C_6H_2O_6 \ + \ x_4O_2 \quad \text{(R)}$$

In order to balance the chemical equation, we have to choose x_1, x_2, x_3 and x_4 such that both sides have the same number of carbon atoms on each side, the same number of hydrogen atoms on each side, and

the same number of oxygen atoms on each side.

An example of the system of linear equations obtained for balancing a chemical equation is the following :

$$x_1C + x_2O \longrightarrow x_3CO_2 \quad \text{(R)}$$

corresponding system of linear equations is:

$$\begin{aligned} x_1 &= x_3 \\ x_2 &= 2x_3 \end{aligned}$$

Consider the system of linear equations obtained for balancing the chemical equation (R) and choose the set of correct options.

- [ ] The nullity of the matrix corresponding to this system is 1.
- [ ] The nullity of the matrix corresponding to this system is 2.
- [ ] $\{(6,6,1,6)\}$ is a basis of the null space of the matrix corresponding to this system.
- [ ] $\{(1,1,6,1)\}$ is a basis of the null space of the matrix corresponding to this system.
- [ ] $\{(1,1,1,6),(6,6,6,1)\}$ is a basis of the null space of the matrix corresponding to this system.
- [ ] $\{(6,6,0,6),(0,0,1,0)\}$ is a basis of the null space of the matrix corresponding to this system.
- [ ] There are infinitely many ways to balance the chemical equation (R).

2.1. Answers :

- The nullity of the matrix corresponding to this system is 1.
- $\{(6,6,1,6)\}$ is a basis of the null space of the matrix corresponding to this system.
- There are infinitely many ways to balance the chemical equation (R).

2.2. Solution

From equation (R) given in the question, we get the following equations : –

Carbon	$x_1 = 6x_3$ $x1 - 6x_3 = 0$
Hydrogen	$2x_2 = 12x_3$ $x_2 - 6x_3 = 0$

Oxygen	$$\begin{aligned} 2x_1 + x_2 &= 6x_3 + 2x_4 \\ 2x_1 + x_2 - 6x_3 - 2x_4 &= 0 \end{aligned}$$

The matrix form $Ax = b$ for the above equations is as follows : –

$$A = \begin{bmatrix} 1 & 0 & -6 & 0 \\ 0 & 1 & -6 & 0 \\ 2 & 1 & -6 & -2 \end{bmatrix},\ x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix},\ b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The reduced row echelon form of $(A|b)$ is given below : –

$$R = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & (-1/6) \end{bmatrix}$$

The first three columns of R have a leading entry, whereas the 4th column of R does not have any leading entry. Therefore, x_4 is an independent variable. Since we have only one independent variable, the nullity of the matrix corresponding to this system of linear equations is 1. Hence, **option1** is correct and **option 2** is wrong.

Let $p = \begin{bmatrix} 6 \\ 6 \\ 1 \\ 6 \end{bmatrix}$ and $q = \begin{bmatrix} 1 \\ 1 \\ 6 \\ 1 \end{bmatrix}$. We can see that $Ap = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ and $Aq = \begin{bmatrix} -35 \\ -35 \\ -54 \end{bmatrix}$. p belongs to the null space of A,
whereas q does not belong to the null space of A. $\{(6, 6, 1, 6)\}$ is also an independent set. The cardinality of this set is 1. We know that the dimension of the null space is also 1. Hence, **option 3** is correct and **option 4** is wrong.

Option 5 and **option 6** are also wrong because nullity is 1. The basis set for the null space can have only one vector.

Let us augment matrix A and b.

$$(A|b) = \left[\begin{array}{cccc|c} 1 & 0 & -6 & 0 & 0 \\ 0 & 1 & -6 & 0 & 0 \\ 2 & 1 & -6 & -2 & 0 \end{array}\right]$$

The reduced row echelon form of $(A|b)$ is given below.

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & (-1/6) & 0 \end{array}\right]$$

As we saw earlier, x_4 is an independent variable. The above is a homogeneous system of linear equations, thereby ensuring the consistency of the system. A system of linear equations will have infinite solutions if it is consistent and has at least one independent variable. Hence, **option 7** is correct.

3. Question

Choose the set of correct options.

- [ ] The null-space of a scalar matrix is always the zero vector space.
- [ ] The null-space of a non-zero scalar matrix is always the zero vector space.
- [ ] The nullity of a non-zero scalar matrix is always 0.
- [ ] The rank of a non-zero scalar matrix is always same as the order of the matrix.
- [ ] The rank of a non-zero scalar matrix is always 0.

3.1. Answers :

- The null-space of a non-zero scalar matrix is always the zero vector space.
- The nullity of a non-zero scalar matrix is always 0.
- The rank of a non-zero scalar matrix is always same as the order of the matrix.

3.2. Solution

Option 1 is incorrect. The nullspace of this scalar matrix $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is $\mathbb{R}^3$.

Option 2 is correct and **option 3** are correct. The reduced row echelon form of any non-zero scalar matrix is the identity matrix. The solution for $I_{n\times n} \cdot x = 0$ is unique, and it is the origin. The dimension of $span\{(0, 0, \ \cdots \ , 0)\}$ is 0. Here, $I_{n\times n}$ is a $n \times n$ identity matrix.

Option 4 is correct and **option 5** is incorrect. Let A be a non-zero scalar matrix.
By Rank-Nullity Theorem, $rank(A) + nullity(A) = \ number\ of\ columns\ in\ matrix\ A.$
Since $nullity(A) = 0$, $rank(A) = number\ of\ columns\ in\ matrix\ A$

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4. Question

Let $A \in M_{5\times 3}(\mathbb{R})$ be a non-zero matrix. Select all true statements from the following:

- [ ] The nullity of A can be at most 2.
- [ ] The nullity of A has to be at least 2.
- [ ] The rank of A can be at most 3.
- [ ] The rank of A has to be at least 3.

4.1. Answers :

- The nullity of A can be at most 2.
- The rank of A can be at most 3.

4.2. Solution

The rank of the matrix A is the number of linearly independent columns of A. Matrix A has 3 columns. There can at most be three linearly independent vectors in the columns of A. Therefore, the rank of A can be at most 3. Since, A is a non-zero matrix, there will be at least one linearly independent column in matrix A. Therefore, the rank of A should at least be 1. Hence, **option 3** is correct and **option 4** is incorrect.

By Rank-Nullity Theorem, $rank(A) + nullity(A) = \ number\ of\ columns\ in\ matrix\ A$.
If $rank(A) = 3$, we get $nullity(A) = 0$. If $rank(A) = 1$, then $nullity(A) = 2$. So, the nullity of matrix A can at most be 2. But we can see that the nullity of matrix A can be less than 2. Hence, **option 1** is correct and **option 2** is incorrect.

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5. Question

Which of the following maps are linear transformations?

- ☐ $T : \mathbb{R} \to \mathbb{R}^2$ defined by $T(x) = (3|x|, 0)$.
- ☐ $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x, y) = (6x + 1, 2x, 2x + 3y + 6)$.
- ☐ $T : \mathbb{R}^2 \to \mathbb{R}$ defined by $T(x, y) = 2x - 4y$.
- ☐ $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x, y) = (4x + 8y, 6x - 3y)$.

5.1. Answers :

- $T : \mathbb{R}^2 \to \mathbb{R}$ defined by $T(x, y) = 2x - 4y$.
- $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x, y) = (4x + 8y, 6x - 3y)$.

5.2. Solution

A linear mapping $T : \mathbb{R}^m \to \mathbb{R}^n$ is a linear transformation if and only if it satisfies the following two conditions for any $v_1, v_2 \in \mathbb{R}^m$ and for any $c \in \mathbb{R}$: –
• $T\ (v1\ +\ v2)\ =\ T\ (v1)\ +\ T\ (v2)$
• $T\ (cv1)\ =\ cT\ (v1)$

Option 1 is incorrect. let $v_1 = 1$ and $v_2 = -1$.

$$LHS \implies T(v_1 + v_2) \implies T(1 + (-1)) = T(0) = (3|0|, 0) = (0, 0)$$

$$RHS \implies T(v_1) + T(v_2) \implies T(1) + T(-1) = (3|1|, 0) + (3|-1|, 0) = (3, 0) + (3, 0) = (6, 0)$$

$LHS \neq RHS$

Option 2 is incorrect. let $v_1 = (1, 0)$ and $v_2 = (1, 1)$.

$T(v_1 + v_2)$	$T(v_1) + T(v_2)$
$= T((1,0) + (1,1))$ $= T(2,1)$ $= (6(2) + 1, 2(2), 2(2) + 3(0) + 6)$ $= (13, 4, 10)$	$= T(1,0) + T(1,1)$ $= (7, 2, 8) + (7, 2, 11)$ $= (14, 4, 18)$

$LHS \neq RHS$

Option 3 is correct. Let $v_1 = (a, b)$, $v_2 = (p, q)$ and $c \in \mathbb{R}$. Here, $a, b, p, q \in \mathbb{R}$.

$T(v_1 + v_2)$	$T(v_1) + T(v_2)$
$= T((a,b) + (p,q))$ $= T(a + p, b + q)$ $= 2(a + p) - 4(b + q)$ $= 2a + 2p - 4b - 4q$	$= T(a,b) + T(p,q)$ $= 2(a) - 4(b) + 2(p) - 4(q)$ $= 2a + 2p - 4b - 4q$

$LHS = RHS$

$T(c \cdot v_1)$	$c \cdot T(v_1)$
$= T(c \cdot (a,b))$ $= T(ca, cb)$ $= 2(ca) - 4(cb)$ $= 2ca - 4cb$	$= c \cdot T(a,b)$ $= c(2(a) - 4(b))$ $= 2ca - 4cb$

$LHS = RHS$. Therefore, $T(x, y) = 2x - 4y$ is a linear transformation.

Option 4 is correct. Let $v_1 = (a, b)$, $v_2 = (p, q)$ and $c \in \mathbb{R}$. Here, $a, b, p, q \in \mathbb{R}$.

$T(v_1 + v_2)$	$T(v_1) + T(v_2)$
$= T((a,b) + (p,q))$ $= T(a + p, b + q)$ $= (4(a + p) + 8(b + q), 6(a + p) - 3(b + q))$ $= (4a + 4p + 8b + 8q, 6a + 6p - 3b - 3q)$	$= T(a,b) + T(p,q)$ $= (4a + 8b, 6a - 3b) + (4p + 8q, 6p - 3q)$ $= (4a + 8b + 4p + 8q, 6a - 3b + 6p - 3q)$

$LHS = RHS$

$T(c \cdot v_1)$	$c \cdot T(v_1)$
$= T(c \cdot (a, b))$ $= T(ca, cb)$ $= (4(ca) + 8(cb), 6(ca) - 3(cb))$ $= (4ca + 8cb, 6ca - 3cb)$	$= c \cdot T(a, b)$ $= c(4a + 8b, 6a - 3b)$ $= (4ca + 8cb, 6ca - 3cb)$

$LHS = RHS$. Therefore, $T(x, y) = (4x + 8y, 6x - 3y)$ is a linear transformation.

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6. Question

Consider a map $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$ and $A = \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix}$. Which of the following options are correct?

- ☐ T is neither one-one nor onto.
- ☐ T is a linear transformation.
- ☐ T is one-one but not onto.
- ☐ T is both one-one and onto.

6.1. Answers :

- T is a linear transformation.
- T is both one-one and onto.

6.2. Solution

Let $v_1 = (a, b)$ and $v_2 = (l, m)$, where $a, b, l, m \in \mathbb{R}$. Let $c \in \mathbb{R}$.

$T(v_1 + v_2)$	$T(v_1) + T(v_2)$

$= T((a,b)+(p,q))$ $= T(a+p, b+q)$ $= \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} a+p \\ b+q \end{bmatrix}$ $= \begin{bmatrix} 17a+17p+b+q \\ 20a+20p+6b+6q \end{bmatrix}$	$= T(a,b)+T(p,q)$ $= \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} p \\ q \end{bmatrix}$ $= \begin{bmatrix} 17a+b \\ 20a+6b \end{bmatrix} + \begin{bmatrix} 17p+q \\ 20p+6q \end{bmatrix}$ $= \begin{bmatrix} 17a+b+17p+q \\ 20a+6b+20p+6q \end{bmatrix}$

$LHS = RHS$

$T(c \cdot v_1)$	$c \cdot T(v_1)$
$= T(c \cdot (a,b))$ $= T(ca, cb)$ $= \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} ca \\ cb \end{bmatrix}$ $= \begin{bmatrix} 17ca+cb \\ 20ca+6cb \end{bmatrix}$	$= c \cdot T(a,b)$ $= c \cdot \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix}$ $= c \cdot \begin{bmatrix} 17a+b \\ 20a+6b \end{bmatrix}$ $= \begin{bmatrix} 17ca+cb \\ 20ca+6cb \end{bmatrix}$

$LHS = RHS$. Therefore, $T(v) = Av$ is a linear transformation. Hence, **option 2** is correct.

A linear transformation is one-to-one if and only for $T(v) = 0$ implies $v = 0$. Let us find $T(v)$ from the given information.

$$T(v) = Av = \begin{bmatrix} 17 & 1 \\ 20 & 6 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 17x+y \\ 20x+6y \end{bmatrix}$$

Let us equate $T(v)$ to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

$$\begin{aligned} 17x+y &= 0 \quad (1) \\ 20x+6y &= 0 \quad (2) \end{aligned}$$

If this system of linear equations is unique, we can say that T is one-to-one.
From (2) we get, $y = -10x/3$. Let us substitute this in (1),

$$\begin{aligned} 17x + (-10x/3) &= 0 \\ (51x - 10x)/3 &= 0 \\ 41x &= 0 \end{aligned}$$

$$x = 0$$

On substituting $x = 0$ in $y = 10x/3$, we get $y = 0$. We have a unique solution for $T(v) = 0$. $T(v) = 0$ implies $v = (0, 0)$, thereby making T one-to-one. Hence, **option 1** is incorrect.

We can see that T is from $\mathbb{R}^2$ to $\mathbb{R}^2$. For T to be onto, for every $w \in \mathbb{R}^2$, there must exist a $v \in \mathbb{R}^2$ such that $T(v) = w$. Let $w = \begin{bmatrix} p \\ q \end{bmatrix}$, where $p, q \in \mathbb{R}$. We know that $T(v)$ is of the form $\begin{bmatrix} 17x + y \\ 20x + 6y \end{bmatrix}$. By equating $T(v)$ to w, we get:

$$\begin{aligned} 17x + y &= p \\ 20x + 6y &= q \end{aligned}$$

T will be onto if there is at least one solution for the above system of linear equations. The coefficient matrix of the above system of linear equations is A. Notice the observation below:

$$det(A) = (17 \times 6) - (1 \times 20) = 102 - 20 = 82$$

Since $det(A) \neq 0$, the solution for the above system of linear equations is unique. Therefore, T is onto. Hence, **option 3** is incorrect and **option 4** is correct.

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7. Question

Consider the matrix A given as follows: $A = \begin{bmatrix} 5 & 3 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 5 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Which of the following sets is a possible basis for the null space of A?

- ☐ $\{(0, 1, 0, 0), (0, 0, 0, 1)\}$
- ☐ $\{(0, 0, 1, 0)\}$
- ☐ $\{(0, 0, 1, 0), (0, 0, 0, 1)\}$
- ☐ $\{(0, 0, 0, 1)\}$

7.1. Answer :

- $\{(0, 0, 0, 1)\}$

7.2. Solution

$Null\ Space\ of\ A = \{(x_1, x_2, x_3, x_4) | x_1, x_2, x_3, x_4 \in \mathbb{R}\ and\ Ax = 0\}$.The reduced row echelon form of $(A|0)$ is $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$. There is no leading entry in 4th column. Therefore, x_4 is an

independent variable. x_1, x_2 *and* x_3 are dependent variables. Therefore, $nullity(A) = 1$. We can see that $x_1 = 0,\ x_2 = 0$ *and* $x_3 = 0$. So, the solution for $Ax = 0$ is of the form $(0, 0, 0, x_4)$.

Option 1 and **option 3** are incorrect as $nullity(A) = 1$, but the cardinality of the set given in these two options is 2. **Option 2** is incorrect as $(0, 1, 0, 0)$ is not of the form $(0, 0, 0, x_4)$.

Option 4 is correct as $(0, 0, 0, 1)$ is of the form $(0, 0, 0, x_4)$, where $x_4 = 1$.

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8. Question

Let A be a 4×3 matrix and let B be a 3×4 matrix. If the null space of A and B are 0 and 4 dimensional vector spaces, respectively, then what is the maximum possible rank for AB?

8.1. Answer : 0

8.2. Solution

It is given that $nullity(B) = 4$. We can see that B has 4 columns. The reduced row echelon form of B will have no leading entries. This is possible only if all the entries of B are equal to 0. Therefore, the rank of AB will always be 0.

Comprehension $(9, 10, 11)$

Shubham, Poulami, Asrifa, Raphael, and Vicky bought the same type of pen drives from a shop. The shopkeeper asked them to give ratings on the quality of the pen drives, 1-star being the lowest denoting $not\ satisfied$ and 5-stars being the highest denoting $extremely\ satisfied$. Whenever a customer gives a rating, say 3-stars rating, then the vector we consider corresponding to that rating is $(0, 0, 1, 0, 0)$. Note that one customer can rate only once and the same rating can be given by more than one customer.

Example: Table $M2W5G2$ shows an example of the rating given by the customers mentioned above as: Shubham gave 2-stars rating, Poulami gave 1-star rating, Asrifa gave 1-star rating, Raphael gave 5-stars rating, and Vicky did not give any rating.

	ratings				
Customers	1-star	2-stars	3-stars	4-stars	5-stars
Shubham	0	1	0	0	0
Poulami	1	0	0	0	0
Asrifa	1	0	0	0	0
Raphael	0	0	0	0	1
Vicky	0	0	0	0	0

Table: M2W5G2

Based on the above information, answer the following questions.

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9. Question

Choose the set of correct options from the following.

- [ ] If Shubham and Poulami both gave 2-stars rating, Asrifa and Raphael both gave 1-star rating, and Vicky gave 3-stars rating, then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 3.
- [ ] If Shubham and Vicky both gave 3-stars rating, Poulami gave 2-stars rating, Asrifa gave 5-stars rating, and Raphael gave 1-star rating then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 4.
- [ ] If Shubham gave 5-stars rating, Poulami gave 3-stars rating, Asrifa gave 1-star rating, Raphael gave 2-stars rating, and Vicky gave 4-stars rating then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 5.
- [ ] If Shubham, Poulami, and Asrifa, all gave 2-stars rating, and Raphael and Vicky gave 1-star rating then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 1.

9.1. Answers :

- If Shubham and Poulami both gave 2-stars rating, Asrifa and Raphael both gave 1-star rating, and Vicky gave 3-stars rating, then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 3.
- If Shubham and Vicky both gave 3-stars rating, Poulami gave 2-stars rating, Asrifa gave 5-stars rating, and Raphael gave 1-star rating then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 4.
- If Shubham gave 5-stars rating, Poulami gave 3-stars rating, Asrifa gave 1-star rating, Raphael gave 2-stars rating, and Vicky gave 4-stars rating then the dimension of the vector space spanned by the row vectors corresponding to the ratings is 5.

9.2. Solution

OPTIONS	MATRIX	REDUCED ROW ECHELON FORM
Option 1 is correct. There are three non-zero rows in the reduced row echelon form, thereby spanning a three dimensional vector space.	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
Option 2 is correct. There are four non-zero rows in the reduced row echelon form, thereby spanning a four dimensional vector space.	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
Option 3 is correct. There are five non-zero rows in the reduced row echelon form, thereby spanning a five dimensional vector space.	$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
Option 4 is incorrect. There are two non-zero rows in the reduced row echelon form, thereby spanning a two dimensional vector space.	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

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10. Question

Consider the matrix A corresponding to the ratings given in Table $M2W5G2$, which is given as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Which of the following statements about the matrix A are true?

- ☐ The set $\{(1,0,0,0,0),(0,1,0,0,0),(0,0,0,1,0)\}$ spans the nullspace of A.
- ☐ The set $\{(0,0,1,0,0),(0,0,0,1,0)\}$ spans the nullspace of A.
- ☐ Nullity of A is 2.
- ☐ Nullity of A is 3.

10.1. Answers

- The set $\{(0,0,1,0,0),(0,0,0,1,0)\}$ spans the nullspace of A.
- Nullity of A is 2.

10.2. Solution

Let us solve $Ax = 0$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$.

The reduced row echelon form for $(A|0)$ is $\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$.

There are no leading entries in the 3rd and 4th columns of the coefficient matrix of the reduced row echelon form of $(A|0)$. So, x_3 and x_4 are independent variables. $Nullity(A) = 2$. Hence, **option 3** is correct and **option 4** is incorrect.

On converting the reduced row echelon form of $(A|0)$ into linear equations, we get:

$$\begin{aligned} x_1 &= 0 \\ x_2 &= 0 \\ x_5 &= 0 \end{aligned}$$

x_3 and x_4 can take any values. So the solution for $Ax = 0$ is of the form $x = \begin{bmatrix} 0 \\ 0 \\ x_3 \\ x_4 \\ 0 \end{bmatrix}$.

Option 1 is incorrect as $(1, 0, 0, 0, 0)$ and $(0, 1, 0, 0, 0)$ are not of the form $(0, 0, x_3, x_4, 0)$.
Option 2 is correct as $(0, 0, 1, 0, 0)$ and $(0, 0, 0, 1, 0)$ are of the form $(0, 0, x_3, x_4, 0)$.

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11. Question

What will be the rank of the matrix A defined in the above question?

11.1. Answer : 3

11.2. Solution

$$\begin{aligned} Rank(A) + Nullity(A) &= Number\ of\ columns\ in\ A \\ Rank(A) + 2 &= 5 \\ Rank(A) &= 3 \end{aligned}$$